

MOUHSSINE EL BOUMSHOULI

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AI Engineering Student - Full-Stack & AI Software Developer

Profile

AI engineering student at UEMF building production-style software across full-stack systems, AI-assisted workflows, computer vision, and data-intensive applications. Hands-on with **FastAPI**, **React**, **PostgreSQL**, **Docker**, computer vision, and human-in-the-loop AI, with a strong focus on testing and deterministic backend logic.

Education

Engineering Degree in Artificial Intelligence – EIDIA, Euromed University of Fes *2025 – Expected 2028*
Fes, Morocco

Preparatory Cycle in Digital Engineering – EIDIA, Euromed University of Fes *2023 – 2025*
Fes, Morocco

Selected coursework: Data Structures & Algorithms, OOP, Databases, Machine Learning, Probability & Statistics, Cybersecurity.

Experience

Bounaim Auto – Software & AI Engineering Intern *Jul – Aug 2026*
Beni Mellal, Morocco

- Independently designed and delivered **SmartImport**, a full-stack procurement decision-support platform for supplier quotations, multicurrency cost analysis, document ingestion, and human-reviewed AI assistance.
- Implemented deterministic **Decimal** financial rules, completeness-aware supplier comparison, staged PDF/XLSX import, **PostgreSQL** persistence, and Dockerized delivery.

Convoroute LLC – Founder & Developer *Jun 2026 – Present*
United States (remote)

- Built a B2B offering for embedded AI chatbots supporting customer service, FAQ automation, and lead capture with recurring subscriptions.
- Own product direction, LLM integrations, Stripe payments, pricing, client delivery, and ongoing development of the self-service platform.

Projects

SmartImport – Procurement Decision-Support Platform *2026*
React · TypeScript · FastAPI · PostgreSQL · SQLAlchemy · Alembic · Docker

- Built end-to-end purchasing workflows: requirements, supplier quotations, FX assumptions, landed-cost and cash-out analysis, document staging, immutable snapshots, and PDF/XLSX exports.
- Validated with **139 backend tests** and synthetic evaluation results of **98.99% document-field F1**, **100% product-line F1**, and **92.31% top-1 matching** on eligible unique-match cases.

medskel – Medical Image Skeletonisation *2026*
Python · OpenCV · Shapely · NumPy

- Implemented a polygon/bisector skeletonisation method for 2D medical segmentations, validated with 28 hand-derived tests and 9 reproducible experiments.
- On 28 retinal images, reduced inter-observer vessel-length variation from **9.3% to 3.2%** versus pixel thinning ($p < 0.0001$).

UEMF Présence – Smart Attendance System *2026*
Next.js · React · TypeScript · PostgreSQL · Prisma · Vercel

- Built and deployed a role-based attendance platform with recurring schedules, GPS check-in and **HMAC-SHA256** QR tokens rotating every 10 seconds.
- Added anti-fraud checks, professor review workflows, CSV reporting, and an administration anomaly dashboard.

Intelligent Autocomplete & Query Suggestion Engine *2026*
C · Ternary Search Tree · Benchmarking

- Built a pure-C autocomplete engine with manual memory management, fuzzy lookup, dynamic updates, top-k ranking, CSV export, benchmarking, and memory-profiling modes.

Skills

Programming: Python, TypeScript, C, C++, Java, SQL, R, MATLAB, HTML/CSS

Backend & Data: FastAPI, SQLAlchemy, Pydantic, PostgreSQL, Alembic, Prisma, REST APIs

Frontend: React, Next.js, Vite, Tailwind CSS

AI & Computer Vision: LLM integration, prompt design, OpenAI APIs, OpenCV, NumPy

Tools & Quality: Docker, Nginx, Git/GitHub, Linux, pytest, Vitest, Ruff, mypy, ESLint, Vercel

Languages: Arabic (native), English (C1 - TOEFL iBT 112/120), French (working proficiency), Spanish (beginner)